

QUSSAI ADLBI

Biomedical Engineering Student & Emerging Researcher

1. PERSONAL INFORMATION

- **Full name:** Qussai Adlbi
- **Nationality:** Syrian
- **Date of birth:** 7 February 2003
- **Address:** Hama, Syria (currently studying in Tartus, Syria, under Erasmus+ mobility in Budapest, Hungary, 2025–2026)
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Google Scholar: scholar.google.com/citations?user=nKaW0H4AAAAJ&hl=en

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2. ABOUT ME

I develop and evaluate biosignal-based methods for decoding motor intent across people a model has never seen. My work builds a coherent research programme on reliable cross-subject decoding — sEMG first, then EEG, multimodal fusion, and simulated systems — under one consistent standard of strict subject-independent evaluation, zero calibration, and full statistical reproducibility. I am completing a B.Sc. in Biomedical Engineering (expected September 2027) and completed an Erasmus+ academic mobility term at Pázmány Péter Catholic University, Budapest, including coursework in Introduction to Artificial Intelligence, Neural Interfaces and Prostheses, and Sensor Technologies and Biological Sensing (completed with full marks).

3. WORK EXPERIENCE

Independent Researcher — Human Motor Intelligence Research Programme

2025 – Present

Al-Andalus University for Medical Sciences, Syria, with Pázmány Péter Catholic University, Budapest, Hungary

- **Main activities:** Design, run, and write up Human Motor Intelligence, a six-part research programme on reliable, zero-calibration, cross-subject decoding of motor intent from surface EMG and EEG.
- **Core sEMG series:** designed a strict leave-one-subject-out (LOSO) benchmark across three NinaPro databases (73 subjects) that exposed rest-class metric inflation in standard reporting; diagnosed, via covariance-eigenvalue analysis, the structural reason domain-adaptation methods fail on MiniROCKET sEMG features; currently developing Lite-DAN, a domain-adversarial learned-invariance approach, under a protocol locked before results are read.
- **EEG & multimodal extension:** ran an externally validated, 54-subject zero-calibration EEG motor-imagery study with a full statistical power analysis, and an externally validated 24-subject EEG-EMG fusion study that also surfaced a novel confidence-accuracy miscalibration finding under signal degradation.
- **Research infrastructure:** built and maintain the programme's supporting infrastructure — four open-source research platforms (MyoControl, MyoAdapt, MyoSim, BioSignal-FM) for measurement, adaptation, and simulation, connected through shared contracts and a public provenance and reproducibility register.

Design & Build — Muscle-Tension Measurement Device

2025 – 2026

Al-Andalus University for Medical Sciences, Syria — Applied Engineering Project

- **Main activities:** co-designed and built a non-invasive surface-EMG muscle-tension measurement device (Ag/AgCl electrodes, AD620 instrumentation amplifier, 450 Hz Sallen–Key low-pass and 50 Hz notch filtering, envelope extraction, Arduino Uno acquisition, Python RMS/MAV/peak analysis interface).
- **Outcome:** completed and delivered to the university; final grade 91% (Excellent).

4. EDUCATION AND TRAINING

B.Sc. in Biomedical Engineering

Expected September 2027

Al-Andalus University for Medical Sciences, Tartus, Syria

- **Graduation thesis — ongoing, formal defense pending (2026/2027):** SurgIntent: Intent-Driven Surgical Robotics — a computational research project, currently in progress, testing whether subject-held-out human-state inference can be connected to a deterministic, robotic-style authorisation policy under explicit uncertainty rules, across three deliberately unpooled contexts: (A) a matched multimodal source-state-proxy study on 12 participants (WAY-EEG-GAL, 3,486 trials), currently reaching mean outer balanced accuracy 0.586 and macro-F1 0.555 under audited nested model selection; (B) a surgical-context, EEG-only comparator study (PhysioNet robotic-assisted-surgery task) in which Ridge and CNN models have so far not outperformed a constant baseline; and (C) a deterministic software policy layer replayed on held-out probabilities, currently producing zero modelled false activations at the cost of zero completed task episodes. A full thesis draft and defense-preparation materials are in hand, with work continuing ahead of a scheduled formal defense. Public release of project materials will follow the defense.

Erasmus+ Academic Study Mobility

2025 – 2026

Pázmány Péter Catholic University, Budapest, Hungary

- **Programme:** International Program, study period 2025/26/1 · language of instruction: English
- **Coursework:** Introduction to Artificial Intelligence · Neural Interfaces and Prostheses · Sensor Technologies and Biological Sensing — completed with full marks.

5. PERSONAL SKILLS

Mother tongue(s)

Arabic

Other language(s)

English — working proficiency demonstrated through a full academic term of coursework and assessment conducted entirely in English (Erasmus+ mobility, Pázmány Péter Catholic University). A formal certified test (IELTS/TOEFL) is planned ahead of graduate-programme application deadlines.

Communication and organisational skills

- Sustained a two-institution, two-country supervisory research relationship (Syria and Hungary) across a multi-manuscript research programme.
- Wrote and maintained public documentation, reproducibility reports, and scientific-limitations registers for six research studies and four open-source platforms.
- Managed a multi-track research programme end to end — protocol design, execution, statistical analysis, manuscript writing, and open-source release.

Digital skills

- Programming & analysis: Python, scikit-learn, statistical hypothesis testing (Friedman, Wilcoxon, Holm correction, power analysis), SHAP explainability, Git/GitHub.
- Engineering & design: SolidWorks, AutoCAD, basic FEA, embedded acquisition (Arduino), analogue signal-conditioning circuit design.
- Documentation & writing: LaTeX/Typst, technical and scientific writing, IMRAD manuscript structure, reproducibility and provenance documentation.

6. ADDITIONAL INFORMATION

Manuscripts

- **Rest-Class Metric Inflation in Zero-Calibration Cross-Subject sEMG: A Three-Database LOSO Benchmark Across Intact-Limb and Transradial Amputee Populations:** Manuscript, submitted to a peer-reviewed journal — under editorial review.
- **Cross-Subject sEMG Gesture Recognition with MiniROCKET: A Mechanistic Analysis of Domain Adaptation Failure:** Manuscript, with supervisor prior to journal submission.
- **Strict Zero-Calibration Cross-Subject EEG Motor-Imagery Decoding: A Harmonized External Validation Study on OpenBMI:** Manuscript under supervisory review.
- **Transparent Cross-Subject EEG-EMG Fusion in Motor Imagery: An Amended External Validation on Jeong2020:** Manuscript under supervisory review.

Open-source research software

- **MyoControl:** inspectable EMG analysis and control-support platform.
- **MyoAdapt:** cross-subject sEMG adaptation research infrastructure.
- **MyoSim:** non-clinical simulation and decoder-provenance research environment.
- **BioSignal-FM:** surface-biosignal representation-learning research framework.

All archived with open GitHub repositories and Zenodo DOIs — see github.com/Qussai-BME.

Honours and awards

- **91% (Excellent):** Muscle-Tension Measurement Device, applied engineering project, Al-Andalus University for Medical Sciences.

References

Available on request.